TUTOR GUIDE • LEVEL 1 • TERM 1 • LESSON 4 OF 10

The 100 Square Explorer



Big idea: Patterns live in rows and columns. The 100 square lists 1 to 100 in rows of ten. Going right adds 1, going down adds 10, and counting in 2s, 5s and 10s makes patterns we can predict.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Whole Numbers (Numbers to 100: counting on and back, reading a number square; Number-sequence patterns (counting in 2s, 5s and 10s); Tens and ones: 10 more and 10 less) Strand label for the tutor only. It is not a claim that the program covers the school syllabus, and it must not appear on child or parent pages as MOE-aligned. Term 1 goes beyond the school Primary 1 syllabus on purpose: counting in 2s, 5s and 10s and moving by rows are P1, but the growing-gap and sum-of-the-two-before sequences, interleaved patterns and counting in 3s are not. They sit at Think harder, Go further and the Investigation only; every Start here item is approachable with the 100 square and counters. Do not tell families this is the school syllabus.
Objectives	• Read the 100 square: rows of ten, 1 more across to the right, 10 more one row down. • Show counting in 10s, 5s and 2s on the square, and say what the pattern and the last digits look like. • Use the square to move on by rows and squares (for example 14, down 2 rows and right 3 squares is 37) without counting every square. • Spot the rule in a number pattern, including one where the gap grows by 1 each time, and give a reason.
Joining cold	**Remember warm-up covers:** Count in 10s to 100 with both hands; Count in 5s (one hand at a time) and 2s (pairs); What comes next after a 9, such as 29 then 30. Only counting to 20 and showing a number on fingers. If the child has done the holiday course, counting in 2s, 5s and 10s will feel familiar; if not, the warm-up teaches it with hands and pairs, and the 100 square shows every number in print. Lesson 4 is the safest lesson to miss in Term 1 (plan, Term 1 note): no later lesson depends on it. A child who misses it watches the recording and does workbook pages 1 and 2. Patterns come back in Lessons 16 and 28 of the year.
Child materials	• The kit page: a printed 100 square (page 6 of the workbook). No cutting is needed. • About 12 small counters of one kind: buttons, coins, dried beans, paper clips or small pieces of paper • Two or three colours (pencils or crayons). With only one colour, use dots and crosses • The printed workbook (pages 1 to 5) and the kit page • A pencil, a rubber and spare paper
Tutor materials	• A slide deck and the 100 square on screen • A visible clock No ruler is needed. Nothing needs to be bought: everything can be found at home. No matches, toothpicks or anything sharp. Counters never go near the mouth or ears.
Tiers	Start here (6 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Contest-style move	Used at Think harder only: changing-step sequence (the gap grows by 1); sum of the two numbers before; two rules mixed in one list; numbers in two lists (2s and 5s). Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first, including the colouring.
- Tell parents a week before: print workbook pages 1 to 6 (page 6 is the 100 square, no cutting needed), find about 12 counters and two or three colours. Counters stay away from mouths and younger children.
- Have the slides open and check the 100 square on screen share. Keep a visible clock: the deck has no timer slide.
- Check the recording captures the tutor and the shared screen. Children's video is excluded; any unmuted voice is recorded. Start the recording after the greeting, and avoid first names while it runs.

Safeguarding and recording (this lesson)

- Counters and small things stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp. Keep small things away from younger brothers and sisters.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a board up to the camera. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for up to 90 days. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–4	**Opening**	4	2: 100 Square Explorer, Get ready	3:36
4–10	**Remember warm-up**	6	3: Warm-up: count in tens, Fives and twos, What comes next?	5:04
10–14	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:10
14–28	**Explore**	14	4: Explore: read the square, Explore: cover the tens, Explore: fives and twos, Explore: move the counter	11:08
28–38	**Mini-lesson**	10	7: Across: 1 more, 1 less, Down: 10 more, up: 10 less, Your turn: write the number, Check: write the number, Stripes and last digits, Gaps can grow, Quick-fire: 10 more	7:27
38–42	**Brain break**	4	1: Jump, clap, tap	3:35
42–70	**Workbook**	28	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	26:51
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	5:17
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:30
82–85	**Look-back**	3	1: Look-back	2:43
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:44
	Total	**90**		**74:03**

The script needs 74:03 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 87:46. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 14 (end of sneak-peek), minute 42 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 42, skip in this order: optional

items and extra quick questions first, then the second explore round, then any quick-fire in the mini-lesson, then the optional slide. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 4:00 Opening

script 3:36 of 4:00

0:00	SLIDE	Show slide **100 Square Explorer** (s-title)
0:25	REC	*If you are watching the recording: hello! Get your 100 square, counters, colours, workbook and pencil. Pause or rewind any time.*
0:36	SAY	Hello, inquirers! Please stay muted unless I invite you to speak; give me a thumbs-up if you want to talk. Cameras on if you can, so I can see your page and hands. Cameras off is fine too.
0:54	SAY	This lesson is recorded so that a friend who is away can catch up. The recording has me, the shared screen and anyone's voice that is unmuted. You may mute and keep your camera off.
1:12	SLIDE	Show slide **Get ready** (s-ready)
1:37	SAY	Today we explore the 100 square. Please find your printed 100 square, about twelve counters, your colours and your workbook.
1:47	CHILDREN	Children find their things. They do not need to show them on camera; if it is easy, a child may tilt the screen so the 100 square can be seen. [1:00]
2:47	DO	If you can, greet each child by first name before you start the recording. While it runs, avoid names. If a child has no counters, suggest buttons, coins, dried beans or small pieces of paper. [0:30]
3:17	SAY	No colours? Use a pencil and make dots and crosses. Counters stay on the table, never near the mouth or ears.
3:28	SAY	A warm-up, two puzzles, then we explore with counters, learn some tricks, move, and work in the workbook.
	NOTE	Later lessons of the term skip the routine and rules slides: this lesson only needs a materials check. Say the recording notice in your own words: the recording has the tutor, the shared screen and any unmuted voice. Mention that the 100 square is on the last page of the workbook and that no cutting is needed.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. The first recording of the term shows the full routine.

4:00 – 10:00 Remember warm-up

script 5:04 of 6:00

4:00	SLIDE	Show slide **Warm-up: count in tens** (s-rem1)
4:25	REC	*Recording: join in! Count out loud and show your fingers. Pause if you need longer.*
4:32	SAY	Stand up if you can. Count in tens with me. Show both hands for every ten: ten, twenty, thirty, all the way to one hundred.
4:45	CHILDREN	Count in tens together to 100, both hands up each time. Then back from 100 to 0, whispering, with big hands on 50. [0:50]
5:35	SLIDE	Show slide **Fives and twos** (s-rem2)
6:00	SAY	One hand is five fingers. Count in fives, one more hand each time: five, ten, fifteen, twenty.
6:08	CHILDREN	Count in fives together to 50, showing one more hand each time. [0:40]
6:48	SAY	Socks come in pairs. A pair is two. Count in twos: two, four, six, eight, ten.
6:56	CHILDREN	Count in twos together to 20, tapping the table once for every two. [0:35]
7:32	SLIDE	Show slide **What comes next?** (s-rem3)

7:56	**SAY**	Now you tell me what comes next.
8:00	**ASK**	Ten, twenty, thirty. What comes next? [wait about 10 s for answers]
8:13	**ASK**	Five, ten, fifteen? [wait about 10 s for answers]
8:24	**ASK**	Two, four, six? [wait about 10 s for answers]
8:36	**ASK**	Twenty-eight, twenty-nine. What comes next? [wait about 12 s for answers]
8:50	**ASK**	Thirty-eight, thirty-nine? [wait about 12 s for answers]
	IF STUCK	If a child hesitates on 30 or 40, say the tens word slowly and count up together: twenty-eight, twenty-nine, thirty. A child who is still unsure of tens words will see all the numbers printed on the 100 square in the Explore.
	NOTE	Answers: 40, 20, 8, 30, 40. Children answer with fingers, hands or a whispered number. Nobody writes. Joining cold: all you need is counting to 20 and showing a number on fingers; the rest is taught here.

10:00 – 14:00 Sneak-peek puzzles

script 3:10 of 4:00

10:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
10:25	**REC**	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
10:34	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
10:47	**READ**	Puzzle one. Count in fives. What are the next two numbers? five, ten, fifteen, twenty, then two blanks.
10:57	**CHILDREN**	Children write the next two numbers on their own paper. [0:45]
11:42	**READ**	Puzzle two. Here is part of the 100 square. A counter is on 36. Ravi moves it down one square. Which number is it on now? Write the number. Then write how you know.
12:00	**CHILDREN**	Children write or draw their answer. [0:55]
12:55	**SAY**	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
13:03	**SAY**	Thank you. We will come back to puzzle two in a few minutes.
	NOTE	Puzzle 1 answer: 25 and 30. Puzzle 2 answer: 46, because down one row is 10 more (a row has ten squares). Look for good ideas to praise later (a child who counted along the row, or who said ten more). Do not write scores or keep a list.

14:00 – 28:00 Explore

script 11:08 of 14:00

14:00	**SLIDE**	Show slide **Explore: read the square** (s-explore1)
14:25	**REC**	*Recording: pause now. Put counters on your 100 square as I say, and notice the shapes they make. Press play when you are ready.*
14:37	**SAY**	Look at your 100 square. Put one finger on the number 1. Slide it along the row, saying each number, all the way to 10.
14:50	**CHILDREN**	Children slide a finger along the first row, saying the numbers. [0:20]
15:10	**ASK**	Where is 11? Slide your finger down from the 1 and find it. [wait about 15 s for answers]
15:31	**SAY**	Each row has ten numbers. The next row starts with 11 and ends with 20.
15:38	**ASK**	Find 100. Which corner is it in? [wait about 15 s for answers]
15:57	**SLIDE**	Show slide **Explore: cover the tens** (s-explore2)
16:22	**SAY**	Round one. Count in tens: ten, twenty, thirty. Put a counter on every number you say. Keep going to one hundred.
16:32	**CHILDREN**	Children count in tens and cover 10, 20, … 100 with counters. Tutor scans the cameras. [0:55]
17:28	**ASK**	What do you see? Tell me what shape the counters make. [wait about 35 s for answers]

18:08	**ASK**	Look at the numbers under your counters. What do they all end in? [wait about 20 s for answers]
18:34	**SAY**	Counting in tens makes a straight line down the last column. Every number ends in zero.
18:42	**SLIDE**	Show slide **Explore: fives and twos** (s-explore3)
19:08	**SAY**	Round two. Clear your counters. Count in fives to fifty, and put a counter on each number.
19:16	**CHILDREN**	Children cover 5, 10, ... 50. [0:50]
20:06	**ASK**	What do you see this time? [wait about 30 s for answers]
20:39	**SAY**	Two columns. Look at the last digits: five or zero.
20:44	**SAY**	Round three. Clear again. Count in twos to twenty.
20:48	**CHILDREN**	Children cover 2, 4, ... 20. [0:45]
21:34	**ASK**	What do you see? Which numbers have no counter? [wait about 30 s for answers]
22:08	**ASK**	If we kept counting in twos, would 37 get a counter? Would 60? How do you know? [wait about 30 s for answers]
22:46	**SLIDE**	Show slide **Explore: move the counter** (s-explore4)
23:12	**SAY**	Round four. Clear the square. Put one counter on 34. I call a move, you move the counter and say the number.
23:22	**ASK**	Right one square. What number? [wait about 12 s for answers]
23:37	**ASK**	Down one square? [wait about 12 s for answers]
23:50	**ASK**	Down one square again? [wait about 12 s for answers]
24:04	**ASK**	Left one square? [wait about 12 s for answers]
24:18	**ASK**	Up one square? [wait about 12 s for answers]
24:32	**ASK**	What did you notice? Which way adds 1? Which way adds 10? [wait about 30 s for answers]
	IF STUCK	A child who counts every square to move down: ask them to count how many squares are in one row, then slide the counter and check the numbers.
	IF STUCK	A child with too few counters: they can cover and uncover. A child who has no printed square can read the square on the slide.
	NOTE	Main room only: no breakout rooms in this lesson, so you can see every camera. Answers for round four: 35, 45, 55, 54, 44. Right adds 1, left takes 1, down adds 10, up takes 10. Building with counters is for every child; nobody is named.

28:00 – 38:00 Mini-lesson

script 7:27 of 10:00

28:00	**SLIDE**	Show slide **Across: 1 more, 1 less** (s-m1)
28:25	**REC**	*Recording: keep watching and copy the examples. Rewind if you miss a step.*
28:32	**SAY**	Look at row four. Moving right, the number goes up by 1. Moving left, it goes down by 1.
28:41	**ASK**	What is 1 more than 40? Where does it sit? [wait about 15 s for answers]
29:01	**SLIDE**	Show slide **Down: 10 more, up: 10 less** (s-m2)
29:26	**SAY**	Now look down the column of the 6s: 6, 16, 26, 36. Every step down adds 10, because a row has ten squares. Going up takes 10 away.
29:40	**ASK**	Start at 52. Go down one row. Which number? [wait about 15 s for answers]
30:00	**SLIDE**	Show slide **Your turn: write the number** (s-m3)
30:24	**READ**	Your turn. Write on your paper: 10 more than 36. 10 less than 58. 1 more than 40.
30:34	**CHILDREN**	Children write three numbers, using their square if they want. [0:55]
31:29	**SLIDE**	Show slide **Check: write the number** (s-m3a)

31:54	**READ**	Forty-six, forty-eight, forty-one.
31:56	**SLIDE**	Show slide **Stripes and last digits** (s-m4)
32:21	**SAY**	Three pictures. Counting in tens makes one column. Counting in fives makes two columns. Counting in twos makes stripes. The last digit tells you which list a number is in.
32:36	**ASK**	Would 45 be coloured counting in fives? Would 45 be coloured counting in twos? How do you know? [wait about 30 s for answers]
33:15	**SLIDE**	Show slide **Gaps can grow** (s-m5)
33:40	**SAY**	Not every pattern has the same gap. Look: 1, 2, 4, 7, 11. The gaps are 1, then 2, then 3, then 4. The gap grows by one each time. So the next gap is 5, and the next number is 16.
34:01	**ASK**	What would the gap after that be? [wait about 15 s for answers]
34:19	**SLIDE**	Show slide **Quick-fire: 10 more** (s-m6)
34:44	**SAY**	Quick-fire. I say a number. You say ten more. Twenty-three.
34:49	**ASK**	Forty-seven? [wait about 12 s for answers]
35:02	**ASK**	Sixty-five? [wait about 12 s for answers]
35:14	**ASK**	Ninety? [wait about 12 s for answers]
	NOTE	Key words to use every time: row, column, across, down, ten more, ten less, last digit. Do not teach multiples, odd and even, or times tables. Counting in 3s is for the Investigation only. The growing-gap pattern is a worked example: do not drill it.
	IF STUCK	If a child says 10 more than 36 is 37: ask them to put a counter on 36 and slide it down one row, then say the new number.

38:00 – 42:00 Brain break

script 3:35 of 4:00

38:00	**SLIDE**	Show slide **Jump, clap, tap** (s-break)
38:25	**REC**	*Recording: stand up and join in! Jump for tens, clap for fives, tap for twos.*
38:32	**SAY**	Stand up and get a sip of water. A big jump for every ten, a clap for every five, a tiny tap for every two.
38:45	**CHILDREN**	Call: tens to 50 (jump), fives to 30 (clap), twos to 12 (tap). Then mix: you call 20, 10 or 6 and children choose the move that fits. Count aloud together. [2:50]

42:00 – 70:00 Workbook

script 26:51 of 28:00

42:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
42:25	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
42:34	**READ**	Page 1, Start here. Question 1: count in tens and colour 10, 20, 30, up to 100. Then fill in the blanks. Question 2: count in fives and colour 5, 10, 15, up to 100. Then fill in the blanks.
42:56	**CHILDREN**	Page 1, questions 1 and 2. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [8:20]
51:16	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
51:41	**READ**	Question 1: every number ends in zero, and they make one line going down. Question 2: every number ends in 5 or 0, and they make two lines going down. Mark your own page in another colour.
52:01	**SAY**	After question 2 you may choose: carry on to page 2, or try Think harder. Everyone may choose, at any time.
52:12	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
52:36	**READ**	Page 2. Question 3: write the next numbers in each pattern. Question 4: count in twos and colour 2, 4, 6 and so on to 30. Look at the last digits. Write the different last digits. Question 5: fill in the empty squares a, b and c. Question 6: Siti counts in fives. Which number is wrong? Circle it. Then write what she should have said.

53:12	**CHILDREN**	Page 2, questions 3 to 6. [8:40]
61:52	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
62:18	**READ**	Question 3: ten, twelve. Thirty, thirty-five. Sixty, seventy. Forty-one, fifty-one. Question 4: the last digits are 2, 4, 6, 8, 0. Question 5: a is 34, b is 43, c is 55. Question 6: 26 is wrong, it should be 25.
62:40	**SLIDE**	Show slide **Choose your next page** (s-choice)
63:05	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.
63:21	**MOVE**	Stand and stretch for 30 seconds, then sit down and choose.
63:51	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5, colouring your kit square), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [5:00]
	IF STUCK	Q1 and Q2 (colouring): count in tens or fives aloud and put a counter on each number first, then colour under the counters. Q3: say the numbers aloud with your fingers. Q5: which squares are next to the empty one? Q6: say the fives with a hand each time.
	IF STUCK	A child who is still stuck ticks 'I am still stuck' on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp. You then help in the open session in the next lesson. Never a private message to a child.
	NOTE	Think harder is for the idea taught today in a new setting. Go further is for mastery and extension. The Investigation uses the 100 square on the kit page: colour lightly. Do not announce who is on which page. Slow class: the workbook part needs about 31:20 against 28:00, so Start here may fill most of it. Take the extra minutes from the flex time, do not skip the answer slides, and say that pages 3 to 5 are for home.

70:00 – 76:00 Talk time

script 5:17 of 6:00

70:00	**SLIDE**	Show slide **Talk and explain** (s-talk)
70:25	**REC**	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*
70:32	**READ**	Counting in tens makes one column on the square. Does counting in fives make one column too? Circle Yes or No. Use your 100 square to show why.
70:47	**CHILDREN**	Children put counters on 5, 10, ... 50 again and look at the columns. [0:50]
71:37	**CHILDREN**	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [1:00]
72:37	**ASK**	Who would like to share? Volunteers first. A pass is always fine. [wait about 90 s for answers]
74:13	**SLIDE**	Show slide **Back to the sneak-peek** (s-sneakans)
74:38	**SAY**	Remember puzzle two? A counter on 36 moves down one square. Yes, it lands on 46. Down one row is ten more.
74:49	**ASK**	Did anyone change their mind about puzzle two while we worked? You can just nod. [wait about 20 s for answers]
	NOTE	Model answer: No. Counting in 5s makes two columns, the 5 column and the 10 column, because a row has ten squares and two jumps of five fill one row. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:30 of 6:00

76:00	**SLIDE**	Show slide **Exit ticket** (s-exit)
76:25	**REC**	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:33	**SAY**	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.

76:41	**READ**	E1. Count in fives: thirty-five, forty, forty-five, then two blanks. What are the next two numbers? E2. A counter is on twenty-six. It moves down one row, then right one square. What number is it on after each move? E3. Mei counts in twos: 2, 4, 6, 8 and so on. Will she say twenty-five? Circle Yes or No.
77:13	**CHILDREN**	Children answer E1 to E3 on their own. [2:30]
79:43	**SLIDE**	Show slide **Check the exit ticket** (s-exitans)
80:08	**READ**	Fifty, fifty-five. Thirty-six, then thirty-seven. No: twenty-five ends in 5, and counting in twos ends in 0, 2, 4, 6 or 8.
80:20	**SAY**	Check your own ticket in a different colour. Well done. Put a star by each one you got right.
	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child's result.

82:00 – 85:00 Look-back

script 2:43 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:25	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:30	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick 'I am still stuck', ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
83:03	**CHILDREN**	Children fill in the look-back box on page 5. [1:40]

85:00 – 90:00 Flex time and goodbye

script 0:44 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:25	**SAY**	Well done, inquirers. Next lesson we add numbers by making tens. Please have about 20 small things of two kinds, the kit page cut out, and a pencil and paper. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: what if?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: this square has only 5 numbers in each row. A counter is on 8 and moves down one row. Which number is it on? And what does going down add now?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 13; going down adds 5, the number of squares in a row. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask How do you know? and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	Ali says: when you add 10 to a number, only the tens digit changes. Is that always true? Try 23, 47 and 89 on your square.	*Recording: pause. Add 10 to 23, 47 and 89. Does only the tens digit change?*	Yes for 23, 47 and 89: 33, 57 and 99. The ones digit stays and the tens digit goes up by 1. But 90 plus 10 is 100, where more than one digit changes, and 95 plus 10 would run off the square (105).
Make a puzzle for your grown-up	4 (optional)	Hide a number on the 100 square. Write 3 clues, like: you say it when you count in 5s; it is in the third row; it is not in the 10s. Give them to your grown-up to find the number.	*Recording: pause. Hide a number on the square and write 3 clues for a grown-up to find.*	Many answers are right. Check that the clues fit the hidden number and fit only one number. The example clues (counting in 5s, third row, not in the 10s) give 25 only.
Teach it back	3 (optional)	Pretend a toy has never seen the 100 square. Teach it why counting in 10s makes a straight column.	*Recording: pause. Teach a grown-up or a toy why counting in 10s makes a column.*	Each row has ten squares. Ten more is one whole row further on, which is the square straight below. So the numbers 10, 20, 30 line up in the last column.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 10 to 14)

Puzzle 1 (access): Count in 5s. What are the next two numbers? 5, 10, 15, 20, __, __.

Answer: 25 and 30

Puzzle 2 (think): This is part of the 100 square. A counter is on 36. Ravi moves it down one square. Which number is it on now? Write how you know.

Answer: 46. Down one row is 10 more, because a row has ten squares.

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **Colour 10, 20, 30 and so on to 100 as you count in 10s. Fill in the blanks.**

(a) Every number I coloured ends in the digit ___.
(b) The colours make a line that goes ___ (across or down).

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Answer: They end in 0. They make a line going down.

Hint (say this first): Count in tens: 10, 20, 30. Put a counter on each number first, then colour under the counters.
Worked solution: The numbers are 10, 20, 30, 40, 50, 60, 70, 80, 90, 100. Every one ends in 0. Each is 10 more than the one before, which is one row down, so they make a straight line down the last column.
Common mistakes: Colouring every number in the last row (91 to 100).; Saying the line goes across: the colours are one under the other, so it goes down.
Time for a middle child: about 3 min.

S2 **Count in 5s. Colour 5, 10, 15 and keep going to 100. Fill in the blanks.**

(a) Every number I coloured ends in the digit ___ or ___.
(b) The colours make ___ lines going down.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

Answer: They end in 5 or 0. They make 2 lines going down.

Hint (say this first): Count in fives with one hand each time: 5, 10, 15. Put a counter on each number first, or colour as you count.
Worked solution: The numbers are 5, 10, 15, 20, and on to 100. They end in 5 or 0. The 5s sit in the fifth column and the 10s in the last column, so there are two lines going down.
Common mistakes: Skipping a number, such as 35 or 45.; Saying 1 line because the tens are in a line too (there are 2 lines: the 5s and the 0s).
Time for a middle child: about 3.5 min.

S3 **What are the next two numbers in each of these patterns?**

(a) 2, 4, 6, 8, ___, ___
(b) 15, 20, 25, ___, ___
(c) 30, 40, 50, ___, ___
(d) 11, 21, 31, ___, ___

Answer: 10, 12 | 30, 35 | 60, 70 | 41, 51

Hint (say this first): Say the pattern out loud with your fingers. Look at the jump from one number to the next.
Worked solution: (a) Counting in 2s: 8, then 10, 12. (b) Counting in 5s: 25, then 30, 35. (c) Counting in 10s: 50, then 60, 70. (d) Each step is 10 more, one row down: 31, then 41, 51.
Common mistakes: (d) writing 32 and 33: the jump is 10, not 1.; Changing the jump half way along.
Time for a middle child: about 3 min.

S4 **Count in 2s. Colour 2, 4, 6 and so on to 30. Look at the last digits.**

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30

The different last digits are ___, ___, ___, ___, ___.

Answer: 2, 4, 6, 8, 0 (any order)

Hint (say this first): Say the numbers in twos as you colour: 2, 4, 6, 8, 10. Look at the number you land on each time.
Worked solution: The coloured numbers are 2, 4, 6, 8, 10, 12 and on to 30. Their last digits are 2, 4, 6, 8, 0, and then the same five again. The colours make stripes.
Common mistakes: Colouring 1, 3, 5 as well.; Writing 5 or 1 as a last digit.
Time for a middle child: about 3 min.

S5 **Fill in the missing number in each empty square.**

33	a	35	36	37
b	44	45	46	47
53	54	c	56	57

(a) Square a is ___.
(b) Square b is ___.
(c) Square c is ___.

Answer: a 34, b 43, c 55

Hint (say this first): Look at the squares next to the empty one. Right is 1 more, left is 1 less, down is 10 more.
Worked solution: (a) 34 is between 33 and 35. (b) 43 is under 33, which is 10 more than 33, and above 53. (c) 55 is under 45 (10 more) and between 54 and 56.
Common mistakes: Writing 44 for b: 44 is next to 43, not under 33.; Adding 1 when moving down instead of 10.
Time for a middle child: about 3 min.

S6 **Siti counts in 5s: 5, 10, 15, 20, 26, 30, 35. One number is wrong. Circle it.**

(a) She should have said ___.

Answer: 26 is wrong. She should have said 25.

Hint (say this first): Say the fives with a hand each time: 5, 10, 15, 20. What comes after 20?
Worked solution: Counting in fives, 20 is followed by 25, then 30. 26 ends in 6, but numbers counted in fives end in 5 or 0. So Siti should have said 25.
Common mistakes: Circling 30 or 35.; Writing 26 again instead of 25.
Time for a middle child: about 2 min.

Think harder (page 3): the taught idea in a new setting

T1 **Put a counter on 27 on your 100 square. Move the counter down 3 rows. Which number does it land on? Write how you know on the lines.**

It lands on ___.

Answer: 57 (each row down is 10 more, so 3 rows is 30 more)

Hint (say this first): One row down is 10 more. How much do 3 rows add?
Worked solution: Each row down adds 10, so 3 rows add 30. 27 + 30 = 57. Check on the square: 37, 47, 57.
Common mistakes: Writing 30 or 33 (adding 3 instead of 3 tens).; Counting 3 squares along the row.
Time for a middle child: about 4 min.

T2 **A counter is on 14. It moves down 2 rows. Then it moves right 3 squares. Which number does it land on? Write each step on the lines.**

It lands on ___.

Answer: 37 (14, 24, 34, then 35, 36, 37)

Hint (say this first): First do the two rows down. Then move right, one square at a time.
Worked solution: Down 2 rows is 20 more: 14, 24, 34. Right 3 squares is 3 more: 35, 36, 37. It stops on 37.
Common mistakes: Adding 2 and 3 to 14 (19).; Going down by 2 squares instead of 2 rows (16).
Time for a middle child: about 4 min.

T3 **Look at this pattern: 6, 7, 9, 12, 16 and so on. Find the next two numbers.**

(a) The gaps between the numbers are ___, ___, ___, ___.
(b) The next two numbers are ___ and ___.

Answer: The gaps are 1, 2, 3, 4. The next numbers are 21 and 27.

Hint (say this first): Find each gap: 7 take away 6, then 9 take away 7. What do the gaps do?
Worked solution: The gaps are 1, 2, 3, 4: each gap is 1 bigger. The next gap is 5, so 16 + 5 = 21. The gap after is 6, so 21 + 6 = 27.
Common mistakes: Adding 4 each time (20, 24).; Finding the gaps but forgetting to use them to go on.
Time for a middle child: about 5 min.

T4 **Mei counts in 2s up to 30: 2, 4, 6, 8 and so on. Ali counts in 5s up to 30: 5, 10, 15, 20 and so on. Write the numbers they both say.**

They both say ___, ___ and ___.

Answer: 10, 20 and 30

Hint (say this first): Write Mei numbers and Ali numbers in two lists. Circle the numbers that are in both lists.
Worked solution: Mei says 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30. Ali says 5, 10, 15, 20, 25, 30. Both say 10, 20 and 30.
Common mistakes: Writing only 10.; Including 5, 15 or 25: Mei does not say numbers ending in 5.
Time for a middle child: about 5 min.

Go further (page 4): mastery and extension

G1 **Here is a number chain: 2, 6, ___, 14, 22, ___, 58. One rule makes the whole chain.**

(a) The two blanks are ___ and ___.
(b) I found the first blank by ___.

Answer: 8 and 36 (2 + 6 = 8, 14 + 22 = 36, and 22 + 36 = 58)

Hint (say this first): Look at 2, 6 and the number after them. What can you do with two numbers to get the next one? Check it on 14, 22 and 58.
Worked solution: Each number after the first two is the two numbers before it added together. 2 + 6 = 8, so the first blank is 8. Check: 6 + 8 = 14 and 8 + 14 = 22. The last blank is 14 + 22 = 36, and 22 + 36 = 58 matches the end of the chain.
Common mistakes: Adding 6 each time (8, 14, 20).; Filling the first blank but not the last.; Not checking the rule on the numbers at the end of the chain.
Time for a middle child: about 5 min.

G2 **Look at this list: 5, 50, 10, 45, 15, 40 and so on.**

(a) The next two numbers are ___ and ___.
(b) The 10th number in the list is ___.

Answer: The next two numbers are 20 and 35. The 10th number is 30.

Hint (say this first): Look at every second number on its own. First: 5, 10, 15. Second: 50, 45, 40.
Worked solution: Numbers 1, 3, 5 go up by 5: 5, 10, 15, then 20. Numbers 2, 4, 6 go down by 5: 50, 45, 40, then 35. The list goes on 20, 35, 25, 30. The 10th number is 30.
Common mistakes: Treating the list as one pattern (adding or taking away the same amount).; Getting the 8th number right and the 10th wrong.
Time for a middle child: about 6 min.

G3 **Look at this block of 4 squares.**

(a) Yellow: 34 + 45 = ___.
(b) Other two: 35 + 44 = ___.
(c) Are the two totals the same? Circle Yes or No.
(d) Now try 52, 53, 62, 63. 52 + 63 = ___ and 53 + 62 = ___.
(e) Does it work every time? Circle Yes or No. Then write why.



> **Answer: Both add up to 79, and 52 + 63 and 53 + 62 both make 115. It always works: 1 and 10 are added in both pairs.**

Hint (say this first): Try your own block of four on the square. Add the two numbers on one slant, then on the other slant.
Worked solution: If the top left number is a, the others are a + 1, a + 10 and a + 11. One slant is a + (a + 11) and the other is (a + 1) + (a + 10). Both are the same: 2 a and 11. For the first block both are 79.
Common mistakes: Only trying one block and saying it always works.; Adding the numbers across the rows instead of along the slants.
Time for a middle child: about 7 min.

Investigation (page 5): Counting in 3s

Easy first step: Use the 100 square on your kit page. Colour 3, 6, 9, 12 and 15. Look at where the colours are.
Then: Keep going to 60, in 3s. Do the colours make a column like the 10s? Write what shape you see. Then look at 33, 36 and 39. What do you notice?

> **Model answer: Counting in 3s makes slanting lines going down and to the left, not a straight column, because a row has 10 squares and 10 is not said when we count in 3s. The picture repeats every 3 rows: 33, 36 and 39 sit under 3, 6 and 9.**

Look for: A child who colours carefully, says what shape they see and gives a reason that uses rows of ten. Any clear noticing counts (zig-zag, slanting, it moves left each row). Colouring to 60 shows the repeat. Do not name the word multiple.
Common mistakes: Colouring a number that is not counted in 3s (say the numbers out loud first).; Expecting a straight column like the 10s.

Talk and explain (minutes 70 to 76)

Counting in tens makes one column on the square. Does counting in fives make one column too? Circle Yes or No. Use your 100 square to show why. Sentence starter: I think ... because ...

> **Model answer: No. Counting in 5s makes two columns, the 5 column and the 10 column. A row has ten squares, so two jumps of five fill one row and the next jump starts the next row in the 5 column.**

Look for: A reason that uses the row of ten (not just because I saw it on my square). Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Count in 5s. What are the next two numbers?**

35, 40, 45, ___, ___

> **Answer: 50, 55**

Worked solution: Counting in fives: 45, then 50, then 55.
Common mistakes: Writing 46 and 47.; Counting in tens (55 and 65).

E2 **A counter is on 26 on your 100 square. First it moves down 1 row. Then it moves right 1 square.**

(a) After going down it is on ___.
(b) After going right it is on ___.

> **Answer: 36, then 37**

Worked solution: Down one row is 10 more: 36. Right one square is 1 more: 37.
Common mistakes: Writing 27 for down.; Writing 46 (adding 20).

E3 **Mei counts in 2s: 2, 4, 6, 8 and so on. Will she say 25?**

Circle Yes or No.

> **Answer: No. 25 ends in 5. Counting in 2s ends in 0, 2, 4, 6 or 8.**

Worked solution: Counting in twos, the last digit is always 0, 2, 4, 6 or 8. 25 ends in 5, so Mei does not say 25.
Common mistakes: Saying yes because 25 is next to 24 or 26.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (finding my place on the square; counting in 2s, 5s or 10s; moving down a row; the pattern where the gap grows (Think harder); something else); *What helped me* (the 100 square or counters; drawing it; trying again; I asked for help; I was not stuck); *Next time I will try* (drawing it; using counters; checking my answer; asking a question); *I am still stuck (I will ask in the class chat or show my tutor on camera)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	Count in 5s. Write the next three numbers: 40, 45, 50, ___, ___, ___.	55, 60, 65
H2	Start here	On the 100 square a counter is on 58. It moves down 1 row. Which number is it on?	68
H3	Start here	(a) Count in 2s: 2, 4, 6, 8, ___ (b) If you keep going, will you say 17? Circle Yes or No.	10. No: 17 ends in 7, and counting in 2s ends in 0, 2, 4, 6 or 8.
H4	Think harder	What is the next number? 6, 8, 11, 15, 20, ___.	26 (the gaps are 2, 3, 4, 5, then 6)
H5	Go further	Find a calendar. Pick 3 dates that have another date straight below them. (a) How much more is the date straight below each one? (b) Is it always the same? Write why.	7 more each time. A row on a calendar is one week of 7 days, like a row of ten on the 100 square.

Online practice (`out/l04/practice`, key `t1l04`): 14 items; 10 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to notice patterns in numbers and to say why. This week: counting in 10s, 5s and 2s on the 100 square, and moving by rows and squares.

Term line (plan, section 5): Your child is learning to see numbers as tens and ones, to notice patterns and to say why. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–4	**100 Square Explorer** s-title	TODAY WE WILL: / Explore the 100 square and find the patterns hiding in its rows and columns.	Lesson 4 of Term 1, 90 minutes. Objective: read the 100 square, notice what counting in 10s, 5s and 2s looks like on it, and use moves across and down. Time plan: opening 0 to 4, warm-up 4 to 10, sneak-peek 10 to 14, explore 14 to 28, mini-lesson 28 to 38, brain break 38 to 42, workbook 42 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Tell the group the lesson is recorded (the tutor, the shared screen and any unmuted voice) and that a child may keep the camera off.
2	0–4	**Get ready** s-ready	Your 100 square (the kit page) / About 12 counters, all one kind / Two or three colours / Workbook, pencil and paper / Counters stay on the table, never near the mouth or ears.	Materials check, about one minute. Every child needs the printed 100 square (workbook page 6), about twelve counters of one kind (buttons, coins, dried beans, paper clips or small pieces of paper), something to colour with, the workbook, a pencil and paper. A child with no colours can make dots and crosses. Check each camera shows the square and the counters. Do not name a child who is missing something; say it to the whole group and offer a substitute.
3	4–10	**Warm-up: count in tens** s-rem1	Show both hands for every ten / 10 20 30 40 50 60 70 80 90 100 / Now count back from 100. Whisper, and shout on 50!	Count in tens to 100 with both hands up on every number, then back to 0. This is a spiral review of skip counting (a holiday topic) and lets a new child join: they hear it and see it before they say it. Two hands are ten fingers, so each new number is ten more. Do not write anything.
4	4–10	**Fives and twos** s-rem2	One hand is five fingers. Count in fives: / 5 10 15 20 25 30 / Socks come in pairs. A pair is two. Count in twos: / 2 4 6 8 10 12	Fives: show one hand, then two hands, then three hands (one hand and the next), saying five, ten, fifteen. Twos: hold up two fingers for each pair of socks. For a child who only counts to 20, this is the bridge: the numbers are the same, only the jump is bigger. Everyone says the numbers out loud; nobody writes.
5	4–10	**What comes next?** s-rem3	10 20 30 __ 5 10 15 __ 2 4 6 __ 28 29 __ 38 39 __ / Say the next number. Show it on your fingers if you like.	Answers: 40, 20, 8, 30, 40. The last two questions show that after a 9 the next ten begins (29 then 30): a new ten, in the same row of the 100 square. Say the tens word slowly. If a child hesitates, count up from 20 together. Hands up on camera is fine; no marks.
6	10–14	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / Count in 5s. What are the next two numbers? / 5 10 15 20 __ __ / PUZZLE 2: think: / This is part of the 100 square. A counter is on 36. / Ravi moves it down one square. / Which number is it on now? Write how you know. / [picture: hundred]	Puzzle 1 (5, 10, 15, 20, then 25 and 30) is an access question: any child who can count on can try it, with fingers if they like. Puzzle 2 is a thinking question on the lesson key idea: down one row is 10 more. The row under 31 to 40 is hidden on purpose. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera: children keep their papers and check their own thinking at Talk time. There is no private chat and no board in front of the group. Answers: 25 and 30; 46 (down one row is 10 more). Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.
7	14–28	**Explore: read the square** s-explore1	[picture: hundred] / Put a finger on 1. Slide it along to 10. / Where does 11 sit? Slide your finger down. / Find 100. Which corner is it in?	Round 0, about 3 minutes, objects and fingers first. Every child has the printed 100 square. They slide a finger along the first row (1 to 10), then discover that 11 sits directly under 1 and that each row has ten numbers. 100 is in the bottom right corner. This orients a child who has never seen the square. If a child has no printed square, they can look at this slide.

8	14–28	**Explore: cover the tens** s-explore2	[picture: hundred] / Count in tens: 10, 20, 30, … / Put a counter on each number you say. / What do you see? / SAY IT: / The counters make a line that goes ___ (across or down).	Round 1, about 4 minutes. Children count in tens to 100 and put a counter on every number they say. Ten counters make one straight line down the last column. Ask what the numbers under the counters end in: every one ends in 0. Repeat each child answer in your own words so the recording picks it up. Counters can be buttons, coins or beans; a child with fewer can cover and uncover.
9	14–28	**Explore: fives and twos** s-explore3	[picture: hundred] / Clear your counters. Count in fives to 50. / What do you see? / Clear again. Count in twos to 20. / Which numbers have no counter? / Counting in 2s: will 37 get a counter? Will 60?	Rounds 2 and 3, about 4 minutes. Fives to 50 make two columns (the 5 column and the 10 column); the numbers end in 5 or 0. Twos to 20 make stripes: counters on every other square, ending in 2, 4, 6, 8, 0; the numbers with no counter are the odd ones. Then ask the prediction: if we kept counting in twos, would 37 get a counter? No, it ends in 7. Would 60? Yes. Ask how they know. Do not use the words multiple or odd and even unless a child does.
10	14–28	**Explore: move the counter** s-explore4	[picture: hundred] / Counter on 34. Move it right 1. / Now down 1. Down 1 again. / Left 1. Then up 1. / WHAT DO YOU NOTICE?: / Which way adds 1? / Which way adds 10?	Round 4, about 3 minutes. Everyone puts one counter on 34 and moves it as you call. Right 1 gives 35, down 1 gives 45, down again 55, left 1 gives 54, up 1 gives 44. After each move children say the new number. Then ask what they notice: right adds 1 and left takes 1; down adds 10 and up takes 10. This is the key idea of the lesson; the mini-lesson names it. Main room only so you can see every camera.
11	28–38	**Across: 1 more, 1 less** s-m1	[picture: hundred] / Right: 1 more. Left: 1 less. / 40 is the last number in its row. The next number, 41, starts the next row.	Show rows three to five of the square (21 to 50). Moving right adds 1 and moving left takes 1. 40 is the last number of its row; the count carries on at the start of the next row: 41. Ask: what is 1 more than 40? Answer: 41 (still just 1 more, even though it starts the next row). Keep it to about a minute.
12	28–38	**Down: 10 more, up: 10 less** s-m2	[picture: hundred] / Down: 10 more. Up: 10 less. / A row has ten squares. So one step down is ten more.	Look down the column of the 6s: 6, 16, 26, 36 and on. Each step down adds 10, because there are ten squares in a row. Going up takes 10 away. Ask: start at 52, go down one row, which number? Answer: 62. This is the idea behind the Think harder questions, where several rows are added at once.
13	28–38	**Your turn: write the number** s-m3	Write on your paper. Timer: 1 minute / 1: / 10 more than 36 / 2: / 10 less than 58 / 3: / 1 more than 40 / [picture: hundred]	Each child writes three numbers on their own paper, using the square if they want. Walk the cameras. Do not ask children to hold boards up in front of the group. Question 3 is the row-end case: 40 is the last number of its row and 41 starts the next.
14	28–38	**Check: write the number** s-m3a	1: / 10 more than 36 / 46 / 2: / 10 less than 58 / 48 / 3: / 1 more than 40 / 41 / [picture: hundred]	Answers: 46, 48, 41. Read each answer aloud so a child watching the recording can mark their own work. For question 3, say that 40 is the end of its row and 41 is the first number of the next row.
15	28–38	**Stripes and last digits** s-m4	[picture: hundred] / [picture: hundred] / [picture: hundred] / THE LAST DIGIT: / Counting in 10s: every number ends in 0. / Counting in 5s: every number ends in 0 or 5. / Counting in 2s: every number ends in 0, 2, 4, 6 or 8.	Show the three pictures side by side. Counting in 10s is one column; counting in 5s is two columns; counting in 2s is stripes, every other square. The last digit tells you which list a number is in. Ask: would 45 be coloured counting in 5s? Yes, it ends in 5. Would 45 be coloured counting in 2s? No. Do not use the words multiple, odd or even yet; say numbers you say counting in 2s.
16	28–38	**Gaps can grow** s-m5	[picture: seq] / The gap grows by 1 each time. So the next gap is 5.	A worked example of a pattern that does not have the same gap each time: 1, 2, 4, 7, 11. Read the gaps: 1, 2, 3, 4. The gap grows by one each time, so the next gap is 5 and the next number is 16. Ask: what would the gap after that be? Answer: 6. This is a worked example only, about a minute; Think harder uses a different pattern of the same kind.
17	28–38	**Quick-fire: 10 more** s-m6	23 47 65 90 / Say the number that is 10 more. Think: one row down. / [picture: hundred]	Say each number and ask for 10 more. Answers: 33, 57, 75, 100. Children answer aloud or with a whispered number; nobody writes. If a child counts on by ones, say: one row down is ten more. For 90, point out that 100 is the last square.

18	38–42	**Jump, clap, tap** s-break	A big jump for every ten. A clap for every five. A tiny tap for every two. / 10 5 2 / Sip some water.	Four minutes. Stand up on camera. Call the pattern and let children move together: tens with a jump to 50, fives with a clap to 30, twos with a tap to 12. Then mix: call a number and children choose the move that fits. No equipment. Keep it gentle for small spaces.
19	42–70	**Workbook page 1: Start here** s-ws1	Questions 1 and 2. Timer: 8 minutes / questions S1, S2	Read both questions aloud. Children colour on the printed squares on page 1, then fill in the blanks. The tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 2 any child may choose to carry on or try Think harder.
20	42–70	**Check page 1** s-ws1ans	answers S1, S2	Read each answer aloud. Children mark their own page in a different colour. Question 1: every number ends in 0 and they make a line going down. Question 2: every number ends in 5 or 0 and they make two lines going down. Never ask a child to show a wrong answer to the group.
21	42–70	**Workbook page 2: Start here** s-ws2	Questions 3 to 6. Timer: 9 minutes / questions S3, S4, S5, S6	Read question 3 and question 6 aloud. Question 4 colours the first three rows of the square in twos. Question 5 has three empty squares labelled a, b and c in a piece of the square. Question 6 is a spot-the-mistake. Children may use their kit square and counters to decide.
22	42–70	**Check page 2** s-ws2ans	answers S3, S4, S5, S6	Read each answer aloud so a child watching the recording can mark their own work. Question 6: Siti said 26, which does not end in 0 or 5; she should have said 25. Never ask a child to show a wrong answer to the group.
23	42–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Our idea in a new setting / Go further (page 4) | Bigger puzzles / Investigation (page 5) | Counting in 3s / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes. The Investigation uses the 100 square on the kit page (page 6), coloured lightly.
24	70–76	**Talk and explain** s-talk	TALK ABOUT IT: / Does counting in 5s make one column, like counting in 10s? / Use the 100 square to show why. / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no. Counting in 5s makes two columns, the 5 column and the 10 column, because a row has ten squares and two jumps of five fill one row. Repeat each child answer in your own words for the recording.
25	70–76	**Back to the sneak-peek** s-sneakans	[picture: hundred] / The counter is on 46. Down one row is 10 more.	Reveal puzzle 2 now: 36 down one square is 46, because each row has ten squares. Say: many of you wrote a good reason. Do not read out anyone earlier answer.
26	76–82	**Exit ticket** s-exit	On your own. No looking back. Timer: 3 minutes / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud (the questions are on the slide). Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
27	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children check their own ticket in a different colour. For question 3 the answer is No. If a child says why, accept any clear reason about the last digit.
28	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.

29	85–90	**Well done, inquirers!** s-end	Show a grown-up one thing you found out today. / NEXT LESSON: / Adding by making tens / Have ready: 20 small things, kit page cut out.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. If the whole class is ahead, show the optional menu slide and use the Spare-time menu from the tutor guide.
30	optional	**Go further: what if?** s-opt	[picture: hundred] / Here is a square with only 5 numbers in each row. / A counter is on 8. It moves down 1 row. Which number is it on? / What does going down add now?	Optional, only if you are on time. This is a square with 5 numbers in each row (1 to 5, 6 to 10, and so on). Answer: 8 down one row is 13, so going down adds 5: the number of squares in a row. Take two answers and move on. This is only the same idea in a new setting; it does not start new content.
31	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Add 10 to a number. Does only the tens digit change? / Make a puzzle | Hide a number on the square and write clues for your grown-up / Teach it back | Show why counting in 10s makes a column / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get your 100 square, counters, colours, workbook and pencil. Pause or rewind any time.*	Gather your 100 square, counters, colours, workbook and pencil. Press play when ready.
4:00	Remember warm-up	*Recording: join in! Count out loud and show your fingers. Pause if you need longer.*	Play along. Count out loud and show your fingers. Pause if you need longer.
10:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
14:00	Explore	*Recording: pause now. Put counters on your 100 square as I say, and notice the shapes they make. Press play when you are ready.*	Pause. Put counters on your square for each pattern and write what you notice.
28:00	Mini-lesson	*Recording: keep watching and copy the examples. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
38:00	Brain break	*Recording: stand up and join in! Jump for tens, clap for fives, tap for twos.*	Stand up and jump, clap and tap with the counting.
42:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*	Pause. Explain your answer to someone at home, with a reason.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one thing you found out.*	Show a grown-up your workbook and tell them one thing you found out.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	4:00	3:36	4:12	OVER (slow class)
remember	6:00	5:04	6:00	OVER (slow class)
sneak	4:00	3:10	3:41	ok
explore	14:00	11:08	13:13	ok
mini	10:00	7:27	8:48	ok
break	4:00	3:35	4:16	OVER (slow class)
work	28:00	26:51	31:59	OVER (slow class)
talk	6:00	5:17	6:17	OVER (slow class)
exit	6:00	4:30	5:17	ok
look	3:00	2:43	3:12	OVER (slow class)
flex	5:00	0:44	0:50	ok
Total	90:00	74:03	87:46	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 6.5 min for page 1 and 11.0 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.